

# Where the wine isn't

## A world map of land fit for vineyards, and what a shrinking market would pay for its wine

*Land fit for vineyards lies idle across Turkey, Ukraine, southern Russia and Iran. We map it at 9km resolution, test the map on continents and countries the model never saw, and price what the land forgoes in a world that drinks less wine every year*

Climate 1991-2020; vineyards as mapped in September 2026; wine-grape area to 2023; market data to 2025

### 70m

Hectares of farmland, not under vines, that pass our strict test of vineyard suitability. The world grows wine grapes on 4.5m

### 0.91

Accuracy (AUC) at telling vineyards from other farmland in countries the model never saw; a textbook temperature rule scores 0.72

### 30% vs 3%

18 countries that plant little of their suitable land: their share of it, against their share of the world's wine grapes

### \$16bn

Wine sales a year those countries forgo against their share of suitable land, at export prices (range \$8bn-27bn)

*And much as Wine has play'd the Infidel,  
And robb'd me of my Robe of Honor—Well,  
I wonder often what the Vintners buy  
One half so precious as the stuff they sell.*

Omar Khayyam, in Edward FitzGerald's version of the Rubaiyat (fifth edition, 1889)

Omar Khayyam, the mathematician and astronomer of Nishapur, is remembered in the West for quatrains in praise of wine, though scholars doubt that he wrote many of the verses that bear his name. Nishapur lies in Khorasan, in north-eastern Iran. By our reckoning Razavi Khorasan has more farmland suited to vines than any other Iranian province, 0.5m hectares on our strict test. Since 1979 Iran has banned alcohol for Muslims, and almost none of that land grows grapes for wine.

Where wine is made owes as much to history as to geography. Monks, merchants, colonists and migrants decided where vines went, and laws and religions decided where they could not. This report asks a narrower question: if wine were grown wherever the land suits it, where would it be grown, and what are the places that grow almost none of it giving up?

We answer in four steps. First, we build a model of what vineyard land looks like, from the climate, water and soil under the vineyards of 29 wine countries, and test it on continents and countries it was not shown. Second, we score every 9km square of the world's land and measure the farmland that looks like vineyard land but is not under vines. Third, we value what the countries that plant least of it forgo, in a world market that has been shrinking since 2018. Fourth, we change every choice in the analysis to see which results survive. A further section asks whether the same land data can predict how good a region's wine is.

This is the second version of the report. A review of the first found that its headline accuracy flattered the model, that its training set left out the hot, dry vineyards of the Mediterranean, and that its measure of suitable farmland counted land already under vines. All three are corrected here; the changes are listed at the end.

## The data

Every input was downloaded, none typed by hand, and all are open. Climate is TerraClimate's monthly normals for 1991-2020, which give temperatures, rain, sunshine, evaporation and the water deficit at about 4km; we average them to a 5 arc-minute grid, about 9km at the equator, which yields 2.1m land cells between 60°S and 72°N. Soil comes from SoilGrids 2.0 (pH, clay, sand, organic carbon, nitrogen, stones and nutrient capacity at 15-30cm). Irrigation comes from the FAO's Global Map of Irrigation Areas. Cropland is the area of all 173 crops in CROPGRIDS for 2020.

The hardest input is the map of today's vineyards. CROPGRIDS includes grapes, but for most countries it inherits a census-based map from 2000 that spreads each district's grapes across the district, or the national total across the cropland: every Turkish province appears to grow grapes on 2-3% of its farmland, every Iranian one on 1-1.4%. It shows no grapes at Stellenbosch or in Marlborough and puts South Africa's vineyards in the Karoo. We use OpenStreetMap instead, the open map of the world, whose volunteers have drawn 872,520 vineyards covering 3.46m hectares, about half the world's vine area. For South Africa it gives 119,887 hectares, against 119,471 from an independent download of the same map. We use OpenStreetMap for a country when it maps at least 35% of the country's grape area, which brings in Spain, Italy and Moldova, and CROPGRIDS only where its sources are fine-grained. Eight wine countries, including Greece and Georgia, meet neither test; they are scored but not used to train.

Wine-grape areas by country for 2000, 2010, 2016 and 2023 come from the University of Adelaide's database by Kym Anderson, Signe Nelgen and Germán Puga; production, prices and consumption from Kym Anderson and Vicente Pinilla's database of global wine markets; and the latest world figures from the OIV's report on 2025. Ratings come from X-Wines, 21m consumer ratings of 100,646 wines from 62 countries.

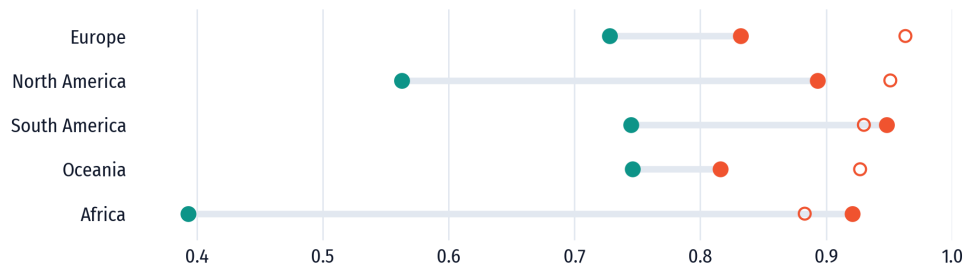
## Reading the land

The score averages two models, each of which learns to tell the 9,073 vineyard cells in the 29 training countries from their other land, using 25 measures of climate, water and soil. One is a set of gradient-boosted decision trees, flexible and best where the training data are rich. The other is the classic tool of ecologists mapping a species' range: a logistic regression on the same measures and their squares, which is rigid but carries over better to places unlike the training data. Neither wins everywhere, so we average their rankings. Two further choices matter. Countries that grow grapes mainly for the table, for raisins or hardly at all are never used as evidence that a place is unsuitable; the absence of vines in Iran or Turkey says more about law, faith and markets than about land. And every training country is scored by models that never saw it, so its scores are tests, not fits.

## Finding vineyards it was never shown

Accuracy at telling vineyards from other farmland on a continent hidden from the model (AUC: 0.5 is a coin toss, 1 is perfect)

Our model Textbook temperature rule



Open circles: the model on all land, including deserts and forests. Asia is left out: only 69 vineyard cells (Armenia) had a trustworthy map. Source: Signum Average

The main test hides a whole continent, and asks the hard version of the question: can the model tell vineyards from the farmland around them? Ruling out deserts, forests and ice is easy and flatters any model; the first version of this report quoted that easier figure. On farmland, the model ranks a hidden continent's vineyard cells above its other farmland with an AUC (the chance that a random vineyard cell outranks a random other cell) of 0.83 in Europe, 0.89 in North America, 0.95 in South America, 0.82 in Oceania and 0.92 in Africa. A textbook rule, which scores land by how close its growing-season temperature is to 17°C, the middle of the 12-22°C band associated with quality wine, manages 0.73, 0.56, 0.75, 0.75 and 0.39. Across all training countries, scored out of country, the model's AUC is 0.91 on farmland and 0.97 on all land, against 0.72 and 0.91 for the rule. Asia cannot be tested: only Armenia's 69 vineyard cells had a trustworthy map.

**Inside each country** Do mapped vineyards score higher than the country's other farmland? AUC, scores from models that never saw the country

|     | Trained on | Vineyard cells | Our model | Temperature rule |
|-----|------------|----------------|-----------|------------------|
| ROU | yes        | 1,117          | 0.66      | 0.61             |
| FRA | yes        | 1,087          | 0.76      | 0.59             |
| ITA | yes        | 766            | 0.55      | 0.49             |
| ESP | yes        | 724            | 0.63      | 0.47             |
| USA | yes        | 314            | 0.88      | 0.48             |
| HUN | yes        | 298            | 0.52      | 0.54             |
| PRT | yes        | 277            | 0.56      | 0.49             |
| CHL | yes        | 242            | 0.78      | 0.61             |
| MDA | yes        | 235            | 0.56      | 0.53             |
| DEU | yes        | 231            | 0.94      | 0.83             |
| AUS | yes        | 230            | 0.84      | 0.74             |
| BGR | yes        | 229            | 0.60      | 0.37             |
| ZAF | yes        | 178            | 0.93      | 0.39             |
| RUS | yes        | 162            | 0.96      | 0.86             |
| ARG | yes        | 151            | 0.76      | 0.53             |
| AUT | yes        | 135            | 0.86      | 0.85             |
| UKR | no         | 133            | 0.90      | 0.76             |
| SVK | yes        | 105            | 0.70      | 0.72             |
| SVN | yes        | 94             | 0.74      | 0.62             |
| CHE | yes        | 63             | 0.73      | 0.71             |
| CZE | yes        | 63             | 0.97      | 0.97             |
| HRV | yes        | 57             | 0.63      | 0.42             |
| SRB | no         | 51             | 0.74      | 0.61             |
| TUR | no         | 46             | 0.39      | 0.30             |

Countries with at least 40 vineyard cells mapped in OpenStreetMap. Source: Signum Average

A harder test still looks inside single countries, where the climate varies less. Here the model is useful but far from perfect: mapped vineyards outscore the rest of a country's farmland with a median AUC of 0.75 in the training countries and 0.72 elsewhere. Countries it never trained on include some of its best results, Ukraine (0.90), Serbia, Algeria and Georgia. Turkey is the exception, at 0.39, and the reason is instructive: 25 of its 46 mapped vineyard cells are in Mardin and Batman, in the hot south-east, where growing-season temperatures exceed 23°C; the model, like the textbook band, rates such heat below the rest of Turkey. Its mapped vineyards elsewhere, in Thrace, the Aegean and Cappadocia, score 0.96 or more. The test says more about where Turkish volunteers map vineyards than about the model, but it means the model's verdict on Turkey, as on Iran and Central Asia, rests on their climates resembling those of proven wine regions rather than on a direct check.

**Known places** Score of the 9km cell at each place and the best cell within 25km; wine-country cells scored by models that never saw that country

|                            | Score of the cell | Best within 25km | Test passed within 25km |
|----------------------------|-------------------|------------------|-------------------------|
| Stellenbosch, South Africa | 0.997             | 0.999            | strict                  |
| Marlborough, New Zealand   | 0.948             | 0.951            | neither                 |
| Central Otago, New Zealand | 0.892             | 0.952            | generous                |
| Napa, United States        | 0.953             | 0.992            | strict                  |
| Willamette, United States  | 0.868             | 0.930            | neither                 |
| Mendoza, Argentina         | 0.968             | 0.970            | generous                |
| Maipo, Chile               | 1.000             | 1.000            | strict                  |
| Barossa, Australia         | 0.989             | 0.999            | strict                  |
| Bordeaux, France           | 0.997             | 0.998            | strict                  |
| Champagne, France          | 0.994             | 0.995            | strict                  |
| Rioja, Spain               | 1.000             | 1.000            | strict                  |
| La Mancha, Spain           | 0.996             | 0.997            | strict                  |
| Mosel, Germany             | 0.989             | 0.990            | strict                  |
| Sussex, England            | 0.958             | 0.963            | generous                |
| Kakheti, Georgia           | 0.985             | 0.999            | strict                  |
| Ningxia, China             | 0.960             | 0.970            | generous                |
| Thrace, Turkey             | 0.987             | 0.995            | strict                  |
| Aegean (Izmir), Turkey     | 0.968             | 0.990            | strict                  |
| Cappadocia, Turkey         | 0.953             | 0.996            | strict                  |
| Shiraz, Iran               | 0.884             | 0.978            | generous                |
| Urmia, Iran                | 0.993             | 0.995            | strict                  |
| Qazvin, Iran               | 0.987             | 0.991            | strict                  |
| Samarkand, Uzbekistan      | 0.995             | 0.995            | strict                  |
| Bekaa, Lebanon             | 0.967             | 0.992            | strict                  |
| Meknes, Morocco            | 0.958             | 0.988            | strict                  |
| Nashik, India              | 0.431             | 0.816            | neither                 |
| Sahara (control)           | 0.604             | 0.724            | neither                 |
| Amazon (control)           | 0.302             | 0.324            | neither                 |
| Siberia (control)          | 0.159             | 0.164            | neither                 |
| Kansas (control)           | 0.855             | 0.905            | neither                 |
| Guangdong (control)        | 0.643             | 0.717            | neither                 |
| Iowa (control)             | 0.693             | 0.803            | neither                 |

Scores are percentile ranks of all land. Strict test: 0.981 or more; generous: 0.951 or more. Source: Signum Average

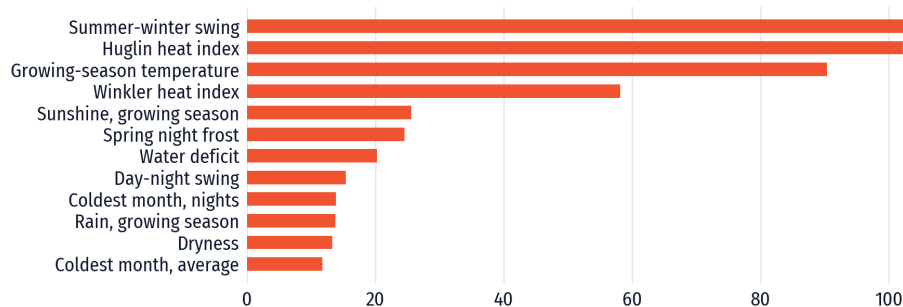
The third test looks up well-known places, taking the best score within 25km so that a single cell cannot miss a district's vineyards. Of 26 wine and grape districts, 18 pass the strict test, among them Stellenbosch, Napa, Maipo, the Barossa, Bordeaux, Champagne, Rioja, La Mancha, the Mosel, Kakheti, Thrace, the Aegean coast, Cappadocia, Urmia, Qazvin and Samarkand. Central Otago, Mendoza, Sussex, Ningxia and Shiraz pass only the

generous test. Marlborough and the Willamette Valley fall just short of it, and Nashik, where vines are pruned twice a year in the tropics, is far outside the model's experience. All six control places, from the Sahara to the Iowa corn belt, fail.

## What makes vineyard land

Loss of accuracy when one factor is scrambled, thousandths of AUC (both models)

■ Climate and water ■ Soil



Factors overlap, so a factor's loss understates its total influence  
Source: Signum Average

What makes vineyard land? Scrambling one measure at a time and watching accuracy fall, the most important are the textbook ones: the gap between summer and winter temperatures, the Huglin and Winkler sums of warmth over the growing season, the growing-season temperature itself, sunshine, and night temperature in April, when vines bud and frost kills shoots. Soil matters little at this scale. Because the measures overlap, each loss understates a factor's full influence.

## The idle acres

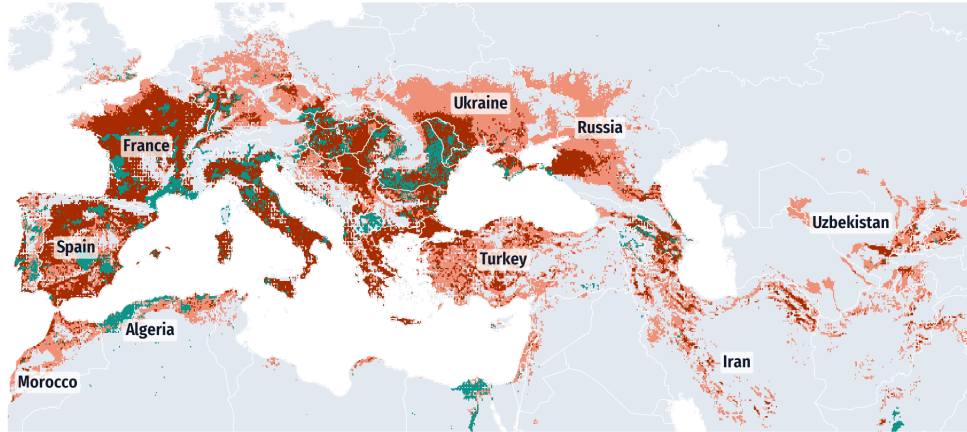
We define two tests. The strict test is passed by land scoring at least as high as the vineyards that hold three-quarters of today's vine hectares in the training countries; the generous test by land as good as those holding nine-tenths. The strict test is the worst case for a newcomer and the generous one the best. Suitable farmland is the cropland in cells that pass and are at least 5% cropland, minus the vines already there.

## Vineyard land without vineyards

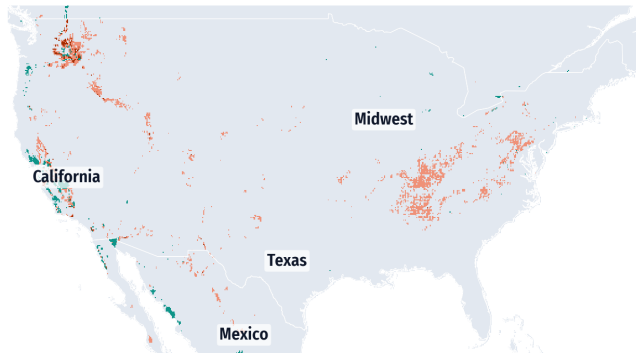
Farmland whose climate and soil match the world's wine regions, by a model that never saw the country it scores. Each cell is about 9 km across; areas are to scale

■ Grapes grown today ■ Suitable, strict test ■ Suitable, generous test

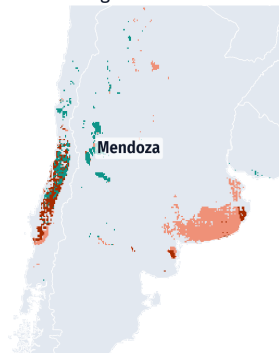
Europe and the Middle East



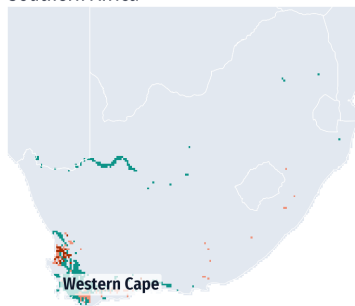
North America



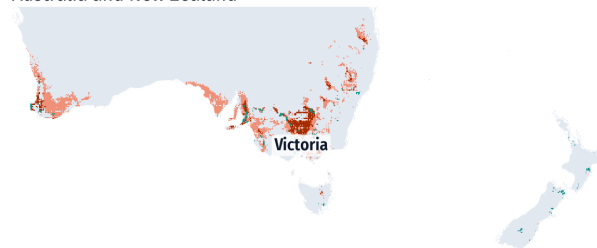
Chile and Argentina



Southern Africa



Australia and New Zealand



Sources: OpenStreetMap contributors; CROPGRIDS (2020); TerraClimate (1991-2020); SoilGrids 2.0; FAO; Natural Earth; The Signum Average

Vineyard land is not scarce. Farmland not under vines that passes the strict test covers 70m hectares, and 159m pass the generous test. The world grows wine grapes on 4.5m. Most of the rest grows other crops, as it should.

**Where the suitable land is** Farmland without vines passing each test, and grapes grown today; thousand hectares

|           | Strict test | Generous test | All grapes, 2020 | Wine grapes, 2023 |
|-----------|-------------|---------------|------------------|-------------------|
| France    | 12,743      | 13,854        | 766              | 858               |
| Spain     | 8,740       | 11,313        | 880              | 928               |
| Romania   | 6,323       | 7,370         | 160              | 178               |
| Italy     | 6,308       | 6,364         | 674              | 650               |
| Turkey*   | 4,473       | 12,303        | 397              | 14                |
| Russia    | 3,997       | 15,859        | 75               | 56                |
| Ukraine*  | 3,483       | 17,625        | 35               | 15                |
| Hungary   | 3,460       | 3,796         | 58               | 56                |
| Bulgaria  | 2,518       | 3,532         | 37               | 54                |
| Greece*   | 2,032       | 2,286         | 97               | 52                |
| Serbia*   | 1,937       | 2,185         | 20               | 6                 |
| Iran*     | 1,780       | 6,757         | 174              |                   |
| Moldova   | 1,685       | 1,715         | 107              | 83                |
| Australia | 1,155       | 6,091         | 142              | 145               |
| Germany   | 1,082       | 5,953         | 101              | 102               |

\*Not used to train the model. Sources: OpenStreetMap; CROPGRIDS; Anderson, Nelgen and Puga; Signum Average

Eighteen countries, each with at least 100,000 hectares of strict-test farmland, plant wine grapes on less than 4.9% of it, the lower quartile of established wine countries. Together they hold 30% of the world's strict-test farmland and 3% of its wine grapes. Turkey has the most: 4.5m hectares on the strict test, 12.3m on the generous one, much of it in Thrace, around Konya and Ankara and in the north-west. It grows grapes on about 400,000 hectares, mostly for the table and for raisins, but wine grapes on only 14,000. Southern Russia (4.0m hectares), around Krasnodar and Stavropol, Ukraine (3.5m) and Serbia (1.9m) follow. Iran grows grapes on 174,000 hectares and has 1.8m hectares of strict-test farmland, most of it in the north-west, around Ardebil, Tabriz, Zanjan and Qazvin, and in Khorasan in the north-east. Algeria and Morocco have let their vineyards shrink: between 2000 and 2023 Algeria's wine grapes fell from 30,200 hectares to 2,300 and Morocco's from 49,600 to 6,700. Uzbekistan, Azerbaijan, Slovakia and the Czech Republic complete the larger gaps.

**The largest unplanted areas** Provinces with under 1% of their strict-test farmland under vines; thousand hectares

|                 | Country    | Strict test | Generous test | Vines today |
|-----------------|------------|-------------|---------------|-------------|
| Krasnodar       | Russia     | 2,298       | 3,116         | 17.6        |
| Vinnytsya       | Ukraine    | 497         | 1,448         | 2.1         |
| Aisne           | France     | 393         | 404           | 3.4         |
| Eure-et-Loir    | France     | 385         | 385           | 0.4         |
| Deux-Sèvres     | France     | 314         | 314           | 3.1         |
| Seien-et-Marne  | France     | 310         | 310           | 1.7         |
| Somme           | France     | 294         | 395           | 0.4         |
| Oise            | France     | 275         | 291           | 0.6         |
| Indre           | France     | 273         | 273           | 2.5         |
| Ille-et-Vilaine | France     | 244         | 244           | 0.5         |
| Meuse           | France     | 221         | 226           | 0.1         |
| Samarkand       | Uzbekistan | 217         | 245           | 0.0         |
| Kherson         | Ukraine    | 216         | 1,111         | 2.1         |
| Adygey          | Russia     | 212         | 230           | 0.4         |
| Chernivtsi      | Ukraine    | 210         | 297           | 0.4         |

Sources: OpenStreetMap; CROPGRIDS; Natural Earth; Signum Average

The largest unplanted province anywhere is Russia's Krasnodar, with 2.3m hectares of strict-test farmland; Ukraine's Vinnytsya and Uzbekistan's Samarkand also stand out. Established wine countries have plenty of unplanted land too, notably the arable north of France. Unlike the first version of this report, the model no longer rates the American Midwest highly: none of it passes the strict test, though parts of Kentucky and Tennessee pass the generous one.

**Who planted, who pulled up** Wine-grape area, thousand hectares, in the first recorded year and in 2023

|                | First year | Area then | Area 2023 | Change |
|----------------|------------|-----------|-----------|--------|
| United Kingdom | 2000       | 0.9       | 4.2       | +382%  |
| New Zealand    | 2000       | 9.9       | 42.0      | +322%  |
| China          | 2010       | 29.5      | 71.2      | +141%  |
| Japan          | 2010       | 3.7       | 6.5       | +74%   |
| Canada         | 2000       | 8.5       | 13.6      | +60%   |
| Czechia        | 2000       | 11.3      | 17.7      | +56%   |
| Israel         | 2000       | 4.9       | 6.5       | +34%   |
| United States  | 2000       | 175.7     | 235.5     | +34%   |
| Spain          | 2000       | 1,181.8   | 928.1     | -21%   |
| Serbia         | 2000       | 69.0      | 6.5       | -91%   |
| Austria        | 2000       | 97.0      | 43.8      | -55%   |
| Romania        | 2000       | 222.2     | 178.2     | -20%   |
| Croatia        | 2000       | 59.4      | 16.2      | -73%   |
| Morocco        | 2000       | 49.6      | 6.7       | -87%   |

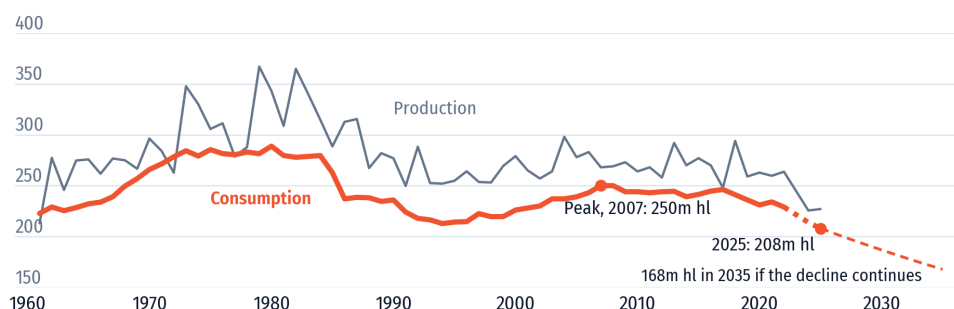
Countries with at least 1,000 hectares in 2023. Countries first recorded after 2000 may have had vineyards earlier. Source: Anderson, Nelgen and Puga (2025)

Where has wine actually spread? England's wine-grape area grew almost fivefold between 2000 and 2023, from 900 hectares to 4,200; New Zealand's quadrupled, to 42,000; China's more than doubled from 2010. Meanwhile Spain pulled up a fifth of its vines, and Serbia, Croatia, Algeria and Morocco most of theirs. The model rates Sussex only on the generous test and Marlborough just below it. Warming may be outpacing a 30-year climate average, and cool, wet vineyards are still a minority in the training data: both point to the model being conservative at the cool edge.

## A shrinking glass

### Drinking less

World wine production and consumption, million hectolitres



2023 not shown: the two sources measure it differently. 2024 derived from the OIV's change for 2025  
Sources: Anderson and Pinilla, University of Adelaide (1961-2022); OIV (2024-25); Signum Average

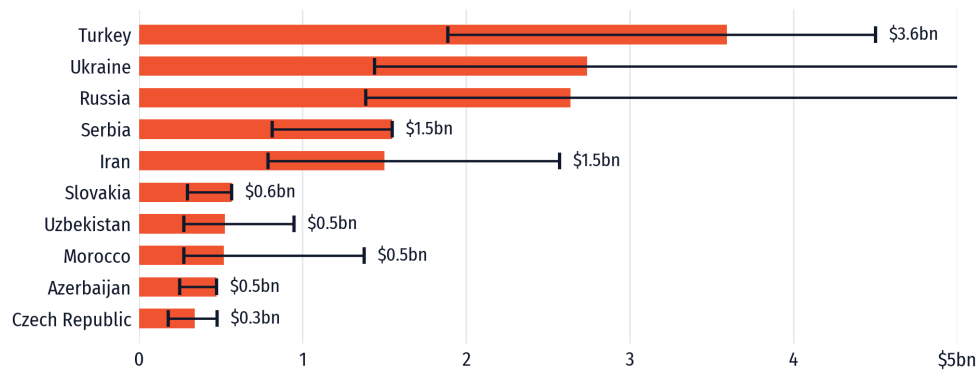
The world drank 208m hectolitres of wine in 2025, according to the OIV, 2.7% less than in 2024 and 14% less than in 2018. In Anderson and Pinilla's series, consumption peaked at 250m hectolitres in 2007 and has never been as low as in 2025 since the series began in 1961. China, France and America account for much of the fall. If the decline of 2018-25, 2.1% a year, continues, the world will drink 168m hectolitres in 2035. Production, at 227m hectolitres in 2025, still exceeds what is drunk; about 30m hectolitres a year go to distilling, vinegar and other uses.

That changes what "forgone" can mean. Planted as densely as the median wine country plants its suitable land, and yielding its median of 57 hectolitres a hectare, the 18 laggards' strict-test farmland would make about 120m hectolitres a year, 58% of world consumption. Land is not the constraint; demand is.

So we measure what these countries miss as a share of the market. Suppose each held a share of world sales equal to its share of the world's strict-test farmland. The 18 would then sell about 55m hectolitres more than they do. At the median export price of established producers in 2018-22, US\$2.84 a litre, that is worth \$16bn a year. In the low case, the market shrinks to its 2035 level and the wine sells at the lower-quartile price of \$1.85: \$8bn. In the high case, each country takes the larger of its shares of strict and generous land: \$27bn.

## The missing vintages

Wine sales a year that each country forgoes against its share of the world's vineyard-suited farmland, US\$bn at export prices; bars are central, lines low to high



Low: strict-test land, 2035 market if the decline continues, lower-quartile price. High: the larger land share, 2025 market  
Sources: OIV; Anderson and Pinilla; Anderson, Nelgen and Puga; OpenStreetMap; Signum Average

## Market share forgone 21 countries that plant wine grapes on under 2.6% of their strict-test farmland

|             | Share of suitable land | Share of wine grapes | Wine forgone, m hl a year | Sales forgone, \$bn a year (low-high) |
|-------------|------------------------|----------------------|---------------------------|---------------------------------------|
| Turkey      | 6.4%                   | 0.30%                | 12.6                      | 3.59 (1.89-4.50)                      |
| Ukraine     | 5.0%                   | 0.33%                | 9.6                       | 2.74 (1.44-6.51)                      |
| Russia      | 5.7%                   | 1.24%                | 9.3                       | 2.64 (1.39-5.30)                      |
| Serbia      | 2.8%                   | 0.14%                | 5.5                       | 1.55 (0.81-1.55)                      |
| Iran        | 2.5%                   | 0.00%                | 5.3                       | 1.50 (0.79-2.57)                      |
| Slovakia    | 1.2%                   | 0.26%                | 2.0                       | 0.57 (0.30-0.57)                      |
| Uzbekistan  | 0.9%                   | 0.00%                | 1.8                       | 0.52 (0.28-0.95)                      |
| Morocco     | 1.0%                   | 0.15%                | 1.8                       | 0.52 (0.27-1.38)                      |
| Azerbaijan  | 0.8%                   | 0.00%                | 1.7                       | 0.48 (0.25-0.48)                      |
| Czechia     | 1.0%                   | 0.39%                | 1.2                       | 0.34 (0.18-0.48)                      |
| Algeria     | 0.6%                   | 0.05%                | 1.1                       | 0.31 (0.16-1.19)                      |
| Bosnia      | 0.4%                   | 0.00%                | 0.8                       | 0.22 (0.12-0.22)                      |
| Kyrgyzstan  | 0.4%                   | 0.00%                | 0.8                       | 0.22 (0.11-0.27)                      |
| Kosovo      | 0.2%                   | 0.00%                | 0.5                       | 0.14 (0.07-0.14)                      |
| Tajikistan  | 0.2%                   | 0.00%                | 0.4                       | 0.13 (0.07-0.25)                      |
| Afghanistan | 0.2%                   | 0.00%                | 0.4                       | 0.11 (0.06-0.47)                      |
| India       | 0.2%                   | 0.07%                | 0.3                       | 0.09 (0.04-0.18)                      |
| Tunisia     | 0.2%                   | 0.12%                | 0.1                       | 0.03 (0.02-0.37)                      |
| All 21      | 29.6%                  | 3.0%                 | 55                        | 15.7 (8.2-27.4)                       |

Sales at export prices. Sources: OIV; Anderson and Pinilla; Anderson, Nelgen and Puga; Signum Average

Three caveats temper these numbers. They are valued at export prices, well below what drinkers pay, so they understate the value of a wine industry but overstate what a new one would earn at first. They are not new money for the world: in a market that shrinks every year, every bottle a newcomer sells is one an established



producer does not. And they are a benchmark, not a forecast. Land is only one reason wine is made where it is; law, faith, war, sanctions, capital, skills and reputation are others, and in most of these countries they are the binding ones. They are also gross: Turkey's and Iran's grapes already earn money as table grapes and raisins, and land turned to wine would stop earning what it earns now.

## Changing every choice

A referee's first question about any model is how much its results depend on the analyst's choices. So we changed every choice that could move them, one at a time: where the strict test is drawn (at 70% or 80% of today's vines rather than 75%), the minimum share of cropland, the vine threshold, which countries train the model (including the first version's narrower set), the weight given to each continent, soil, irrigation, each of the two models alone, and the trees' settings. Across these 15 versions, accuracy on farmland ranges from 0.88 to 0.91; strict-test farmland from 58m to 102m hectares; and sales forgone from \$13bn to \$24bn a year, against \$16bn in the main version. The ranking of countries is stable: its rank correlation with the main version is at least 0.76 and above 0.9 in most versions. Turkey and Ukraine are among the five largest losers in every version, Russia in 14 of the 15 and Iran in 12. Two choices move the results most: where the strict test is drawn, and using the species-distribution model alone, which counts more land as suitable and brings India and Myanmar into the top five.

**Changing every choice** Each row changes one choice; accuracy out of country, and what it does to the results

|                                          | AUC, farmland | Strict-test farmland, m ha | Ranking vs main | Sales forgone, \$bn | Largest five            |
|------------------------------------------|---------------|----------------------------|-----------------|---------------------|-------------------------|
| Base (version 2)                         | 0.906         | 70                         | 1.00            | 15.7                | TUR, UKR, RUS, SRB, IRN |
| Strict test at 70% of vines              | 0.906         | 58                         | 0.98            | 12.7                | TUR, RUS, UKR, SRB, IRN |
| Strict test at 80% of vines              | 0.906         | 98                         | 0.91            | 21.5                | UKR, TUR, RUS, IRN, SRB |
| Farmland threshold 2%                    | 0.906         | 71                         | 1.00            | 15.6                | TUR, UKR, RUS, SRB, IRN |
| Farmland threshold 10%                   | 0.906         | 68                         | 0.99            | 15.9                | TUR, UKR, RUS, SRB, IRN |
| Trees only                               | 0.901         | 68                         | 0.95            | 14.8                | RUS, UKR, TUR, SRB, IRN |
| Envelope only                            | 0.894         | 96                         | 0.76            | 24.1                | TUR, IND, UKR, MMR, RUS |
| Version 1 training set (OSM 50% bar)     | 0.888         | 95                         | 0.88            | 21.8                | TUR, UKR, RUS, IRN, SRB |
| Vine threshold 0.25%                     | 0.905         | 73                         | 0.98            | 16.5                | TUR, RUS, UKR, IRN, SRB |
| Vine threshold 1.00%                     | 0.903         | 80                         | 0.97            | 19.6                | TUR, RUS, UKR, IRN, SRB |
| Continents weighted equally              | 0.895         | 80                         | 0.77            | 18.2                | TUR, IRN, UKR, UZB, SRB |
| No soil                                  | 0.907         | 70                         | 0.93            | 14.8                | UKR, TUR, RUS, SRB, UZB |
| Irrigation counted everywhere            | 0.909         | 74                         | 0.94            | 16.5                | UKR, TUR, RUS, IRN, SRB |
| Irrigation left out                      | 0.880         | 102                        | 0.83            | 21.6                | UKR, RUS, TUR, MAR, DZA |
| Smaller trees, faster learning, new seed | 0.904         | 74                         | 0.98            | 17.4                | TUR, UKR, RUS, IRN, SRB |

Ranking vs main: rank correlation of countries' strict-test farmland with the main version. Sales forgone: central case. Source: Signum Average

## Can the land predict quality?

A map of suitable land would be more useful if it could also say how good the wine would be. To test this we used X-Wines: 21m consumer ratings, on a five-star scale, of 100,646 wines rated between 2012 and 2021. We averaged them by region and measured each region against the average for wines of the same type, because sparkling and dessert wines are rated higher than whites; regions with few ratings are pulled towards zero. We located 923 of the 1,204 regions with at least 300 ratings using OpenStreetMap's geocoder and kept the 694 with vines nearby. Each region gets the climate and soil of its vineyards: the average over the grid squares within its outline, weighted by vine hectares.

## Good land, not necessarily good wine

694 wine regions: consumer rating against the average for the wine's type, stars (up), and the model's suitability score for their land (across). Rank correlation -0.05



Dot size: number of ratings (300 to 1,272k). Predicting ratings from climate and soil beats the naive guess by  $R^2 +0.06$  within countries and  $-0.05$  across them. Sources: X-Wines; Signum Average

**Can the land predict the rating?** Regions' ratings against the average for their wine type, predicted from climate and soil

|                         | Better than naive guess ( $R^2$ ) | Rank correlation | Error, model | Error, naive guess |
|-------------------------|-----------------------------------|------------------|--------------|--------------------|
| <b>Out of country</b>   | -0.05                             | 0.14             | 0.186        | 0.182              |
| <b>Within countries</b> | +0.06                             | 0.30             | 0.175        | 0.180              |

Errors in stars. Naive guess: the average region (out of country) or the country's average (within countries). Source: X-Wines; Signum Average

The answer is no, or very little. Asked to predict regions in countries it had not seen, a model of climate and soil does no better than guessing the average ( $R^2$  of  $-0.05$  against that guess; rank correlation 0.14). Within countries it does slightly better than the country's own average, explaining about 6% of the variation (rank correlation 0.30). A linear model, or only the 423 regions with at least 2,000 ratings, give the same answer:  $R^2$  between  $-0.07$  and  $+0.06$  in every version. Land the suitability model rates highly is not rated more highly by drinkers (rank correlation  $-0.05$ ).

There are two readings, and both are probably right. The measurement is coarse: squares 9km across cannot see the slope, aspect and gravel that distinguish a great site from its neighbour, and consumer ratings mix producer, price and vintage with place. And quality is made as much as found, by producers, investment and reputation. Either way we follow our own rule: a model that cannot beat a naive guess is not used to predict. We do not rank the idle land by expected quality. The map says where vines would grow, not how good their wine would be.

## What changed from the first version

The first version of this report was published on October 2nd 2026. A review found three problems, and this version fixes them. Its headline accuracy, 0.96, counted deserts, forests and ice as easy wins; on farmland the first model scored 0.87, and this one 0.91. Its training set left out Spain, Italy and Moldova, whose vineyards OpenStreetMap maps only partly, and with them most hot, dry Mediterranean vineyards; they are now in, and a second, simpler model is averaged in, which together lifted the hard test in North America from 0.82 to 0.89. And its measure of suitable farmland counted all cropland in passing cells, including land already under vines, against what its methods said; the corrected measure is 70m hectares on the strict test rather than 115m. The sales forgone fell from \$17bn to \$16bn a year, the 21 laggard countries became 18, and Turkey, Ukraine, southern Russia and Iran remain the largest.

## What it means

The map shows where vines could go; it does not show where wine would sell. For most of the countries with the most idle vineyard land the obstacles are not agronomic. Iran bans alcohol for Muslims. Ukraine is at war. Russia is under sanctions. Turkey taxes alcohol heavily and restricts its sale. Algeria and Morocco have been pulling up vines for two decades. Where those obstacles ease, the land is there.

For established producers the finding cuts the other way. Their advantage is not land, which is plentiful, but everything else: names, skills, capital and a market that knows them. In a world that drinks less every year, that advantage matters more, not less.

### What this analysis cannot tell you

The model finds land whose climate suits vines; at this scale climate and soil do not predict how wine is rated. It was not tested in Asia, where Turkey, Iran and Central Asia lie, and within single countries it is less sure than between continents. It uses average climate for 1991-2020 and so lags behind warming; cool regions such as southern England are probably underrated. Cells are about 9km across, too coarse for slopes, aspect and individual sites, which decide much of a vineyard's quality. The vineyard map is good in Europe, the Americas, South Africa and Oceania and thin in Asia, where the model could not be tested. Sales forgone are a benchmark against land, valued at export prices, and ignore the law, religion, capital, skills and reputation that explain much of where wine is made.

## Appendix: data and methods

### The grid

All land between 60°S and 72°N on a 5 arc-minute grid (about 9 km at the equator; 2.1m cells). Climate: TerraClimate monthly normals for 1991-2020, averaged from 1/24 degree. From them we compute the growing-season average temperature (April-October, October-April in the south), Winkler degree-days, the Huglin index with its day-length coefficient, night temperature in the ripening month, the coldest month's mean and night temperature, the hottest month's day temperature, the summer-winter and day-night swings, April night temperature (frost at budburst), rain over the year, the growing season, the two months before harvest and the winter, sunshine reaching the ground, the aridity index (rain over potential evaporation) and the growing-season water deficit. Soil at 15-30cm (SoilGrids 2.0): pH, clay, sand, organic carbon, nitrogen, stones and cation-exchange capacity. Irrigation: FAO's Global Map of Irrigation Areas (v5, about 2005), used only where the aridity index is below 0.65; elsewhere irrigation marks intensive farming of any crop and made maize land look like vineyard land. Cropland: the physical area of all 173 crops in CROPGRIDS (2020), capped at the cell's area.

### Where vineyards are

Vineyards come from OpenStreetMap (landuse=vineyard), read from Overture Maps' release of September 23rd 2026: 872,520 mapped vineyards covering 3.46m hectares, their areas computed on the ellipsoid and summed by cell. For South Africa this gives 119,887 hectares against 119,471 from a direct OpenStreetMap download. A country's vineyards are taken from OpenStreetMap if it maps at least 35% of the country's grape area (version 1 required half; Spain, Italy and Moldova, at 38-45%, now qualify), because the models compare mapped vineyards with the land around them and do not need every vineyard; otherwise from CROPGRIDS if less than a fifth of its grapes there come from the coarse year-2000 census map, which misplaces vineyards (it shows none at Stellenbosch or Marlborough); otherwise the country is not used for training. A cell counts as vineyard if at least 0.5% of its land is under vines.

### The model

Trained on the 29 countries where at least half of the grape area is for wine (Anderson, Nelgen and Puga, 2023 wine-grape area against CROPGRIDS grapes), with at least 1,000 hectares of wine grapes and a trustworthy vineyard map: 9,073 vineyard cells. Countries that grow grapes mainly for the table or raisins, or hardly at all, are never used as evidence against a place; they are only scored. Two models are fitted on all vineyard cells and a random sample of other cells: gradient-boosted trees (scikit-learn HistGradientBoostingClassifier: 300 trees, learning rate 0.06, 31 leaves, L2 penalty 1, balanced classes; 400,000 other cells) and a classic species-distribution model (logistic regression on the standardised measures and their squares; 200,000 other cells). The score is the average of the two models' percentile ranks, itself expressed as a percentile of all land. Each training country is scored by models fitted on the other countries (five folds grouped by country); every other country by the average of the five.

Tests. Hiding one continent at a time, the score separates that continent's vineyard cells from its other farmland (cells with at least 5% cropland) with an area under the ROC curve (AUC) of 0.83 in Europe, 0.89 in North America, 0.95 in South America, 0.82 in Oceania and 0.92 in Africa, against 0.73, 0.56, 0.75, 0.75 and 0.39 for a textbook rule (distance of growing-season temperature from 17°C, the middle of the 12-22°C band in Jones, 2006). On all land, which adds easy cases such as deserts and forests, the AUCs are 0.96, 0.95, 0.93, 0.93 and 0.88 (rule: 0.94, 0.87, 0.75, 0.93 and 0.57). Neither component model wins everywhere: the trees do better in Oceania and Africa, the species-distribution model in the Americas, which is why they are averaged. Across all training countries, scored out of country, the AUC is 0.97 on all land and 0.91 on farmland (rule: 0.91 and 0.72). Asia cannot be tested this way: only 69 vineyard cells, all in Armenia, had a trustworthy map.

Within single countries the test is harder, because climate varies less. On farmland, mapped vineyards outscore the country's other farmland with a median AUC of 0.75 in the training countries and 0.72 in others with at least 25 mapped vineyard cells (Ukraine 0.90, Serbia 0.74, Algeria 0.73, Georgia 0.72). Turkey is the exception at 0.39: 25 of its 46 mapped vineyard cells are in hot Mardin and Batman (growing-season temperature above 23°C), which the model, like the textbook band, rates below the rest; its mapped vineyards elsewhere score 0.96-0.997. Province-level census figures cannot serve as a test for Turkey, Iran or Spain, because CROPGRIDS spreads their national grape area almost evenly across provinces.

Of 26 well-known wine or grape regions looked up by name, 23 pass at least the generous test somewhere within 25km (18 the strict test, including Thrace, Cappadocia, Urmia, Qazvin and La Mancha); Marlborough, Willamette and Nashik pass neither. All six control places (the Sahara, the Amazon, Siberia, Kansas, Iowa, Guangdong) fail both. Scrambling one factor at a time, the largest losses of accuracy come from the summer-winter temperature swing, the Huglin and Winkler heat sums, growing-season temperature, sunshine and April night temperature. Soil matters little at this resolution.

## Strict and generous tests

The strict test is passed by land scoring at least as high as the vineyards that hold three-quarters of today's vine hectares in the training countries (score 0.981, a percentile of all land); the generous test by land scoring as high as those holding nine-tenths (0.951). They cover 3.0% and 6.5% of the training countries' land. Suitable farmland is the cropland in cells that pass and are at least 5% cropland, minus the vines already there (OpenStreetMap vineyards or CROPGRIDS grapes, whichever is larger). Worldwide this is 70m hectares on the strict test and 159m on the generous one, against 4.5m hectares of wine grapes in 2023. Version 1 counted all cropland in passing cells, vines included, against what this note said; correcting it is most of the fall from 115m hectares.

## Quality

Ratings: X-Wines, 21m consumer ratings (one to five stars, 2012-2021) of 100,646 wines. A region's score is the sum over its ratings of (rating minus the average for that wine type), divided by the number of ratings plus 300, so that sparse regions are pulled to zero. Regions with at least 300 ratings were located with OpenStreetMap's Nominatim geocoder within their country (923 of 1,204); the 694 with mapped vines within their outline are used. A region's climate and soil are averages over grid squares within its outline (at least 0.25 and at most 1 degree from its centre), weighted by vine hectares. Gradient-boosted regression trees (200 trees, 8 leaves, at least 20 regions a leaf), weighted by the log of ratings, were tested in eight folds grouped by country against the average region, and in eight random folds against the country's average.  $R^2$  against the naive guess: -0.05 out of country (rank correlation 0.14) and +0.06 within countries (0.30). Ridge regression and the 423 regions with at least 2,000 ratings give  $R^2$  between -0.07 and +0.06. Because no version beats the naive guess by a useful margin, the quality model is not used to rank land.

## Market share and sales forgone

Benchmarks are 20 established wine countries in 2018-22 (Anderson and Pinilla): wine grapes cover 4.9% and 10.9% of their strict-test farmland (lower quartile and median); yields are 35, 57 and 79 hectolitres a hectare; export prices US\$1.85, 2.84 and 3.64 a litre (quartiles). The 18 countries that plant less than the lower quartile, with at least 100,000 hectares of strict-test farmland, hold 30% of the world's strict-test farmland but 3% of its wine grapes. Planted like the median wine country, their land could make about 120m hectolitres a year, more than half of what the world drinks, so capacity is not the constraint. We therefore value what each would sell if its share of the world market equalled its share of suitable farmland, minus what it has now. Low case: strict-test land, the market in 2035 if the 2018-25 decline continues (2.1% a year, to 168m hectolitres) and the lower-quartile price. Central: strict-test land, the 2025 market (208m hectolitres) and the median price. High: the larger of the strict and generous land shares, the 2025 market and the median price. Totals: 45, 55 and 96m hectolitres a year, worth US\$8.2bn, 15.7bn and 27.4bn at export prices. This is a benchmark for land left unused, not a forecast: law, religion, capital, skills and reputation decide where wine is made, and new producers would have to take sales from established ones.

## Robustness

Fifteen versions change one choice each: the strict-test cut-off (70% or 80% of vines), the farmland threshold (2% or 10% cropland), the vine threshold (0.25% or 1%), the version 1 training set (OpenStreetMap bar of 50%), equal weight for each continent, no soil, irrigation everywhere or nowhere, each model alone, and smaller trees with a faster learning rate and a new seed. Accuracy on farmland, out of country: 0.88-0.91. Strict-test farmland: 58m-102m hectares. Sales forgone, central case: \$13bn-24bn a year. Rank correlation of countries' strict-test farmland with the main version: 0.76-1.00. Turkey and Ukraine are in the five largest losers in all 15 versions, Russia in 14 and Iran in 12.

## Main sources

OpenStreetMap contributors via Overture Maps (ODbL); CROPGRIDS v1.08 (Tang et al., 2024); TerraClimate (Abatzoglou et al., 2018); SoilGrids 2.0 (Poggio et al., 2021); FAO Global Map of Irrigation Areas v5 (Siebert et al., 2013); Natural Earth; K. Anderson, S. Nelgen and G. Puga, Database of Regional, National and Global Winegrape Bearing Areas by Variety, 2000 to 2023 (University of Adelaide, 2025); K. Anderson and V. Pinilla, Annual Database of Global Wine Markets, 1835 to 2024 (University of Adelaide, 2025); OIV, State of the World Wine Sector in 2025 (2026); X-Wines (Azambuja, Morais and Filipe, 2023).

The data behind every chart can be downloaded from the data on GitHub, free to reuse with credit.

## Training countries and vineyard sources

OpenStreetMap vineyards (26): Argentina, Australia, Austria, Belgium, Bulgaria, Canada, Chile, Croatia, Czechia, France, Germany, Hungary, Italy, Luxembourg, Moldova, New Zealand, Portugal, Romania, Russia, Slovakia, Slovenia, South Africa, Spain, Switzerland, the United Kingdom and the United States. CROPGRIDS vineyards (3): Albania, Armenia and North Macedonia. Wine countries scored but not trained on, for lack of a trustworthy vineyard map (8): Brazil, Cyprus, Georgia, Greece, Israel, Lebanon, Turkmenistan and Uruguay. A wine country is one where wine grapes (Anderson, Nelgen and Puga, 2023) are at least half of all grapes (CROPGRIDS, 2020) and cover at least 1,000 hectares.

## Choices that were tested and changed

Several versions preceded this one, and each change was made because a test failed. The first used CROPGRIDS for every country's vineyards and WorldClim's 1970-2000 climate; its continent hold-out AUC was 0.56 in Africa and 0.72 in Oceania, because the vineyard map was wrong there. The next used OpenStreetMap and TerraClimate's 1991-2020 climate and included irrigation everywhere; irrigation became its second most important factor and it passed the irrigated maize land of Nebraska and Kansas. Leaving irrigation out failed irrigated Mendoza instead; counting it only in drylands kept the accuracy and fixed both. That version was published on October 2nd 2026. A review then found that its accuracy was quoted on all land rather than farmland (0.96 against 0.87), that it was trained without Spain, Italy and Moldova (OpenStreetMap maps 38-45% of their vineyards, below the 50% bar then used) and so with few hot, dry vineyards, and that its suitable farmland included land already under vines. Version 2 (October 4th) lowers the bar to 35%, averages the trees with a species-distribution model, reports accuracy on farmland and subtracts existing vines. The specification grid in the report shows how each choice moves the results.

## Thresholds and areas

Scores are percentile ranks, not probabilities: the models are trained with balanced classes on a sample of non-vineyard land. Thresholds are therefore set by where today's vineyards fall in the ranking, weighted by vine hectares. Suitable farmland is the cropland in a passing cell with at least 5% cropland, minus the vines already there, so forests, towns, pasture and existing vineyards do not count. Countries' totals sum cells by Natural Earth borders; cells on a border go to one country.

## Reproducing the results

The code that downloads every input, builds the grid, trains and tests the model and draws the charts is kept in the study's private repository; the data, the suitability map (GeoTIFF) and these methods are public. The scripts fix every random seed, so a rerun gives the same numbers. Vineyards are read from Overture Maps release 2026-09-23.1; a later release will differ slightly as volunteers add vineyards.